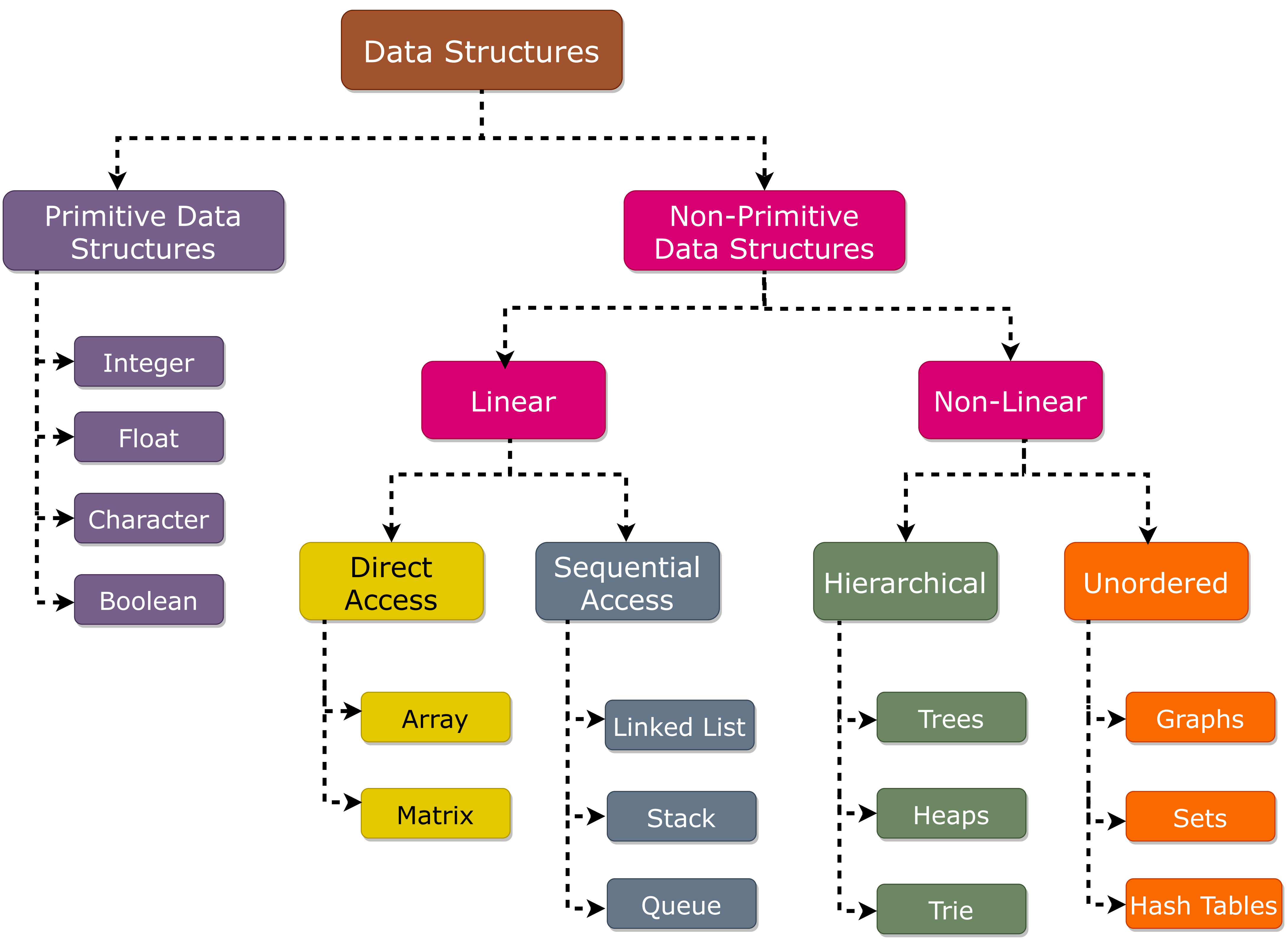
Types of Data Structures

Now that we know what data structures are and why they're important, let's dive into the different types. Data structures can be broadly categorized into:

1. **Primitive Data Structures**: These are basic data structures that include Integers, Float, Character, and Boolean.
2. **Non-Primitive Data Structures**: These are more complex data structures and are further classified into:
    - **Linear Data Structures**: In these data structures, data elements are arranged sequentially. Examples include arrays, linked lists, stacks, and queues.
    - **Non-Linear Data Structures**: Here, data elements aren't placed in a sequence. Examples are graphs and trees.



Types of Data Structures

Let's explore some examples of these data structures.

Primitive Data Structures

Imagine you're making a shopping list. What do you need? Probably fruits, vegetables, snacks, maybe some cleaning supplies. Each of these items can be considered an individual data element - a building block of your list. Primitive data structures in programming are similar. They are the simplest form of data structures and include:

- **Integers**: Think of these as whole numbers, like the number of apples you need to buy.
- **Float**: These are real numbers, like the weight of the bananas you're planning to buy.
- **Character**: A single letter, number, or symbol, like 'A' or '3'.
- **Boolean**: This type holds either a true or false value.

Non-Primitive Data Structures

In contrast, non-primitive data structures are more complex. They are derived from the primitive data structures and can be further divided into linear and non-linear types.

Linear Data Structures

In linear data structures, elements are arranged in a sequential manner. Think of them as a line of people waiting for a bus. Some examples are:

- **Arrays**: Think of an array like a row of lockers. Each locker holds an item and is identified by a unique number.
- **Linked Lists**: Imagine a chain where each link knows about the link following it.
- **Stacks and Queues**: Stacks are like a pile of books, where you can only interact with the top book. Queues, on the other hand, are like a line at a grocery store - the first one in line gets served first.

Non-Linear Data Structures

Non-linear data structures, as the name implies, do not maintain a particular sequence for the arrangement of elements. Examples include:

- **Trees**: Think of a family tree where each person is a node that has connections to their parents and children.
- **Graphs**: These are like social networks, where each person (node) can be connected to multiple other people (nodes).

We've only scratched the surface of data structures here. As you delve deeper into each type, you'll gain a stronger understanding of how they work and where they can be applied.

We will start with the Arrays data structure but before that let's see what is Big-O.

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